Applications of Artificial Intelligence within High Performance Sport

In Partnership With

Global Performance Insights

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Global Performance Insights

This report has been assembled by Jo Clubb, founder of Global Performance Insights in partnership with Zone7 - a proprietary artificial intelligence platform committed to helping sporting organizations unlock greater performance data insights.

Contributors from Zone7 include Tal Brown and Eyal Eliakim, who founded the company together in 2017.

Their experience working within AI at companies like Salesforce and Oracle is tempered with the experiences of Performance Director, Rich Buchanan, former Head of Performance at Swansea City AFC.

Eran Paz, Zone7 Director of Data Science was also brought in to offer his expertise in developing the AI dictionary at the beginning of this report.

Jo Clubb

Founder,
Global Performance Insights

jo.clubb@globalperformanceinsights.com

Sports Science Consultant with more than a decade in high performance sport at teams such as Chelsea FC and the NFL's Buffalo Bills.

Tal Brown

Co-Founder & CEO
Zone7

tal@zone7.ai

A technical leader with more than two decades' experience across product development, and aligning technology solutions with business requirements.

Eyal Eliakim

Co-Founder & CTO
Zone7

eyal@zone7.ai

An experienced Data Scientist and Performance Analyst with a passion for uncovering original insights across the sports industry.

Rich Buchanan

Performance Director
Zone7

rich@zone7.ai

A Chartered Physiotherapist that has held performance roles with the Premier League, Swansea City AFC, FA of Wales, and DC United.

Eran Paz

Director of Data Science
Zone7

eran@zone7.ai

A data science and deep learning researcher and team leader with more than a decade of experience within technology and insurance.

Introduction: Why Dive Deeper into AI in Sport?

Jo Clubb
Global Performance Insights

Late last year, I sought to canvas industry opinion on artificial intelligence (AI) in high performance sport.

In response, I received an array of questions and comments, from both practitioners and researchers all over the globe.

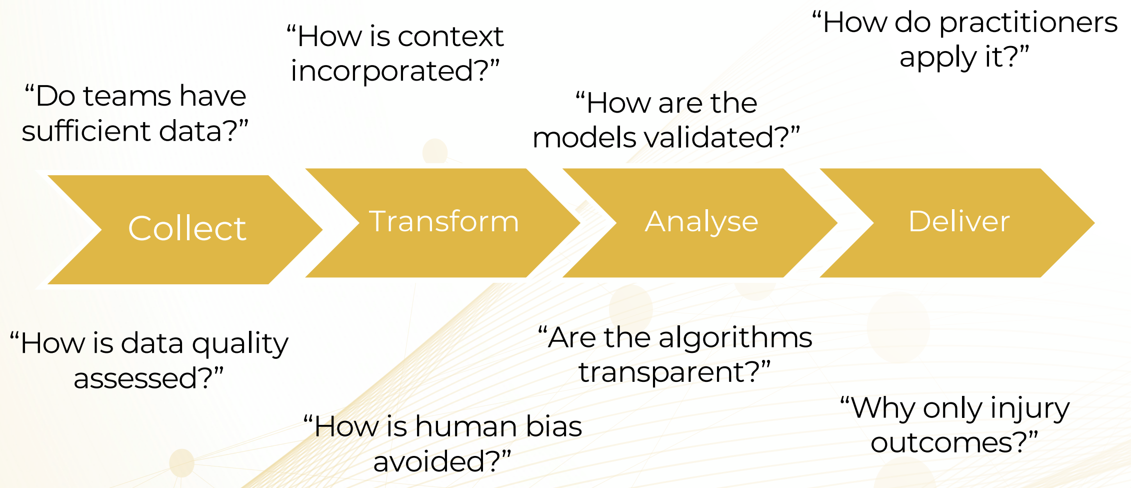
Working closely with Zone7 - a leader in AI application for injury risk forecasting and performance management - this report aims to delve deeper into the key themes that emanated from the discussion.

Questioning The Data Pipeline

Interestingly, there were questions and comments throughout the entire data pipeline, from collection, through transformation, analysis and modelling, to delivery and application.

This reinforces the need to think critically about the entire data process, which is not just specific to AI, but to any time data is collected in sport. Balancing innovation with validation can be a challenge for early adopters, as witnessed with GPS technology uptake in the early 2000s. Yet, as scientists, we should critically evaluate the tools we are using, including statistical and modelling techniques.

Questions throughout the data pipeline



Introduction: Why Dive Deeper into AI in Sport?

Data Quality and Quantity

Another important thread of questions and comments that I received were around data quantity and quality of the inputs going into the model.

This is reflective of a sports science practitioner's process across multiple data streams; "how many trials will provide sufficient quality" or "what is the quality of the tracking data given today's conditions/location" for example. Yet, one thought that strikes me is not "do we have enough", but "*what value can we gain from the data we already have*" ... That said, data quantity in our field warrants much discussion.

Underpowered studies, those with insufficient data points, have been suggested as an issue in sports science (Abt et al., 2020)[1]. As such, aggregated data sets may provide greater insights.

For example, Zone7 Performance Director Rich Buchanan has previously shared on Sportsmith a Zone7 case study[2] in which they performed a retrospective analysis on a dataset from 11 football teams' datasets that included 423 injuries.

As datasets are aggregated, it's important to maintain the context of each specific sport, team, and athlete.

1 https://doi.org/10.1080/02640414.2020.1776002
2 sportsmith.co/articles/artificial-intelligence-in-football-a-new-frontier-for-mitigating-injury-risk/

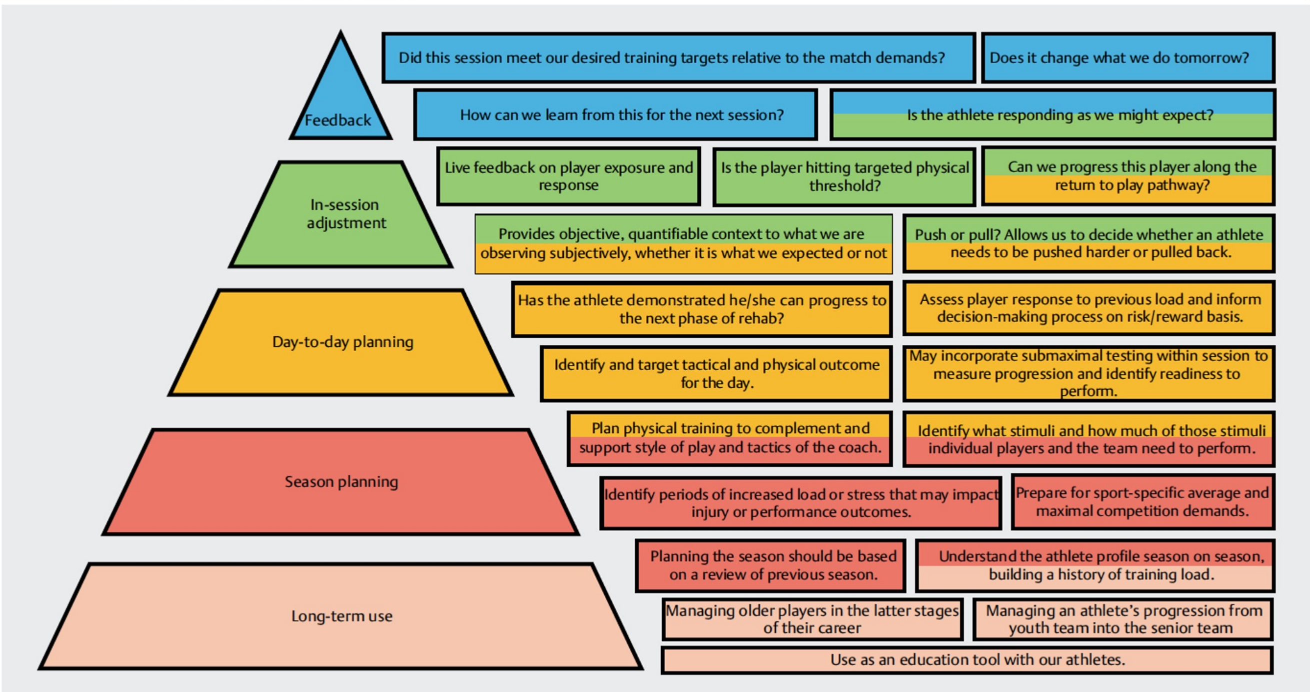


Figure from: West, S. W., Clubb, J., Torres-Ronda, L., Howells, D., Leng, E., Vescovi, J. D., ... & Windt, J. (2021). More than a metric: how training load is used in elite sport for athlete management. *International Journal of Sports Medicine*, *42*(04), 300-306. https://doi.org/10.1055/a-1268-8791

Introduction: Why Dive Deeper into AI in Sport?

... One thought that strikes me is not 'do we have enough?", but '*what value can we gain from the data we already have?*'...

Considering Context

Contextual factors have been identified as a central theme to load management (see figure overleaf; West et al., 2021)[1], so it was not surprising to see context frequently come up in the responses. How can AI techniques incorporate context, such as the playing position, environmental conditions, coach's playing style, time in season, and the different technologies used to capture data, for example.

Modelling Processes

Regarding the modelling process itself, there were comments on transparency. This is a hot topic, especially given the connotations associated with the "black box" approach.

1 https://doi.org/10.1055/a-1268-8791

Like most topics, social media seems to portray this as a binary argument and yet, I believe there must be much greater nuance. Therefore, it is an important theme to explore further.

Applications

Finally, the question of "*so what?*"

What do practitioners do with AI outputs? There are concerns that hark back to Little Britain's catchphrase "*computer says no!*" How should practitioners interact with and apply the outputs? Do they replace or augment decision making? Do they impair or support coach intuition?

Based on these responses, I have extracted the following six key themes:

1. **Definition**: *What does AI and ML mean?*
2. **Inputs**: *Data quantity and quality*
3. **Context**: *Incorporating situational factors*
4. **Transparency**: *Validation of and access to the algorithms*
5. **Outputs**: *Injury risk and load management as outcomes*
6. **Application**: *How do practitioners apply the outputs?*

Definition

What does AI and ML mean?

Definition: *What does AI and ML mean?*

- **General Terminology**
- **Types of Machine Learning Algorithms**
- **Machine Learning Model Attributes**
- **Dataset Types**
- **Algorithm Types**
- **Types of Analysis**
- **Machine Learning Metrics**

This chapter was co-authored by Eyal Eliakim and Eran Paz, Zone7's CTO and Director of Data Science.

Eyal Eliakim
Zone7, Co-Founder & CTO

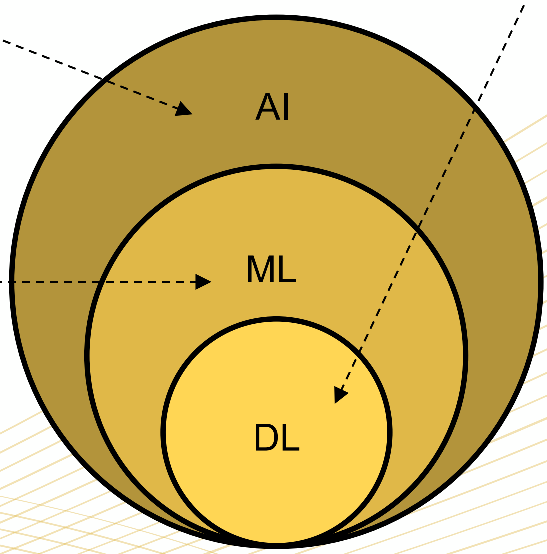
Eran Paz
Zone7, Director of Data Science

General Terminology

Artificial Intelligence (AI) – the ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings. The term is frequently applied to the project of developing systems endowed with the intellectual processes characteristic of humans, such as the ability to reason, discover meaning, generalize, or learn from past experience.

Machine Learning (ML) - is a subset of AI that is particularly focused on developing algorithms that will help machines to perform intellectual processes by learning from historical data and responding to new data.

Deep Learning (DL) – is a subset of ML algorithms that is based on artificial neural networks. Artificial neural networks are able to learn complex tasks in a variety of domains such as computer vision, natural language understanding, speech recognition and many more.



Defining Artificial Intelligence, Machine Learning & Deep Learning

Definition: *What does AI and ML mean?*

Computer Vision (also called Visual Recognition) – is a field of AI that enables computers and systems to derive meaningful information from digital images, videos and other visual inputs — and take actions or make recommendations based on that information. For a recent review of computer vision in sports, see Naik et al., (2022).

Types of Machine Learning Algorithms

Supervised Learning – models are trained to detect underlying patterns between the input data and the labels associated with it, enabling it to yield accurate labelling results when presented with never-before-seen data.

Unsupervised Learning – algorithms discover different patterns and similarities in the data without specifically referencing their relationship with an outcome represented by a label.

Reinforcement Learning (RL) – is an area of ML concerned with how intelligent agents should take action in an environment to maximize the notion of cumulative reward.

RL differs from supervised learning in not needing labelled input/output pairs to be presented and in not needing sub-optimal actions to be explicitly corrected. Instead, the focus is on finding a balance between exploration (of uncharted territory) and exploitation (of current knowledge).

Partially supervised RL algorithms can combine the advantages of supervised and RL algorithms.

Machine Learning Model Attributes

Hyperparameters – are parameters whose values control the learning process and determine the values of model parameters that a learning algorithm ends up learning. They are used by the learning algorithm when it is learning but they are not part of the resulting model.

Each algorithm type can use different types of hyperparameters. Examples for hyperparameters can be the depth of a decision tree, the number of trees in a tree-based ensemble model, and the number of layers in a neural network.

Parameters – are part of the model resulting from the training process. They are learned from the data as the algorithm attempts to map the input features and the labels or targets.

Examples for such parameters are the coefficients of a regression model or the weights of a neural network.

Definition: *What does AI and ML mean?*

Dataset Types

Training Data – is used to train an ML algorithm. For instance, in a supervised learning task, an algorithm looks at the training data (input data and labels) to learn combinations of parameters and values that produce optimal predictive performance. The goal is to produce a fitted model that learns the combinations mentioned above but still generalized and performs well on unseen data.

Validation Data – is a set of examples used to tune a model's hyperparameters and thresholds as well as model selection. It should follow the same distribution as the training data set.

In order to avoid overfitting, when any of a model's attributes needs to be adjusted, it is recommended to have a validation data set. The validation data set is completely separate and does not overlap with the training data set.

Cross Validation – In order to get more stable results and use all valuable data for training, a data set can be repeatedly split into training and validation datasets in multiple iterations. This process is known as cross validation.

The results of the validation process will be determined based on an aggregation of the results achieved in each iteration.

It is common practice to hold out an additional dataset from the cross-validation process as a test set to fully understand a model's expected performance in a real-life scenario.

Test Data – is an independent data set from the training and validation sets, but follows the same distribution as the training data set. Therefore, it is a set of examples used to assess the performance of a model for which all attributes have been selected. The final model selected following training on the training set and tuning on the validation set is evaluated based on it's ability to agree in it's predictions on the test set, when compared to the actual response values linked with it.

Data Lake – is a central location that holds a large amount of data in its native, raw format. Compared to a hierarchical data warehouse, which stores data in files or folders, a data lake uses a flat architecture and object storage, with metadata tags and object storage, to store the data.

Specifically in the context of Zone7, when attempting to create injury risk forecasting models, the data lake represents the dataset used to "train" the models. This is strictly separated from the test data, which is typically an external dataset.

In the case of Zone7's approach, the dataset is divided into seasonal units, where the test data (e.g., the most recent season) is completely separate from the training data (previous seasons).

Definition: *What does AI and ML mean?*

Algorithm Types

Classification – algorithms are a subset of supervised learning algorithms used to identify the category of an unseen observation. A classifier learns from the given dataset and classifies new observations into a number of classes/categories.

For example, Yes or No, 0 or 1, Spam or Not Spam, cat or dog, is an athlete likely to suffer an injury or not. Importantly, the output of such an algorithm can at times be presented as an estimated probability of each class.

Regression – algorithms are supervised learning algorithms that learn a model based on a training dataset to estimate continuous response values.

Types of Analysis

Prospective Data Analysis – is a process that is designed to evaluate a predictive model trained on past data, based on its performance on future events.

For Zone7, this is using an algorithm trained on historical data to forecast the injury risk of players based on new incoming data day-by-day.

Retrospective Data Analysis – is a process that is designed to to train and evaluate a model's performance based on historical data. In order to evaluate the model's performance, the most recent part of the data is held out for evaluation and is treated as "future" relative to the data the model is trained on.

In the context of Zone7, in order to evaluate the system's performance at any given moment in time based on a large sample, using retrospective data in the evaluation data set is necessary. If hypothetically, we have ten seasons of historical data, we can train the system based on athlete data and injuries that occurred in seasons 1-9 and evaluate the system's performance based on its ability to correctly flag injuries that occurred during season 10.

Machine Learning Metrics

Accuracy – of a machine learning system is measured as the percentage of correct predictions or classifications made by the model over a specific data set.

Accuracy is a very simple metric that doesn't perform well when the condition we are trying to predict is very rare. For example, if we would like to train a model that predicts a disease that impacts a single person out of 10,000 people, we can train our model to predict "not sick" for all people who get tested. The model will be correct 9,999 times out of 10,000 which will give it an accuracy of 99.99% but it will be completely useless.

Definition: *What does AI and ML mean?*

In the same manner described above, given an injury is a rare event we (Zone7) could hypothetically train an algorithm to always forecast an injury isn't likely to happen and receive very high accuracy, though as mentioned before, the model will be of no use.

False Negative (Rate) – represents a "positive" case that was incorrectly predicted as "negative".

In the case of injury risk forecasting with Zone7, this would be where a player sustains an injury whilst the system forecasted low injury risk levels preceding it. Another way to think of this is an unexpected injury, where no flag was provided ahead of the incident.

False Positive (Rate) – represents a "negative" case that was incorrectly predicted as "positive". The False Positive Rate represents the proportion of "negative" cases that were incorrectly classified as "positive".

In the case of injury risk forecasting with Zone7, this would be where a player does not sustain an injury, whilst the system forecasted medium/high risk levels.

Sensitivity/ Recall (True Positive Rate) – is the proportion of the correctly classified "positive" cases out of all the "positive" cases. In the case of injury, it's the proportion of injuries for which medium/high risk levels were forecasted prior to occurrence.

Sensitivity reflects the percentage of injuries in a specific dataset that were correctly flagged by the model. For example, 50 injuries were sustained in a season, and this season was used as test data. Results indicate that 41 injuries were correctly flagged. Hence, the sensitivity (also called recall) is 82%. Sensitivity is referred to at times as the injury detection rate.

Specificity (True Negative Rate) – is the proportion of the "negative" class that is correctly classified. In the case of injury, the number of uninjured athletes that are correctly identified.

Overfitting – is a symptom of ML training in which an algorithm tends to "memorize" its training data rather than learning significant patterns from it. This usually leads to inferior performance on new previously unseen data.

Overfitting usually occurs when training data is too small or oversimplified to accurately represent the complexity of the real life situation we are trying to learn.

A few common methodologies to reduce overfitting include:

- Enlarging the training dataset so it's too large to be "memorized":
- Combining the dataset with additional similar datasets
- Artificially augmenting the dataset
- Use simpler models which avoid overfitting but overall tends to have inferior performance

Definition: *What does AI and ML mean?*

Precision – is the proportion of correctly classified "positive" cases out of all the **predicted** "positive" cases.

In the case of injury risk forecasting with Zone7, precision would be the proportion of athletes that were injured out of all the athletes that were forecasted to be at medium/high risk.

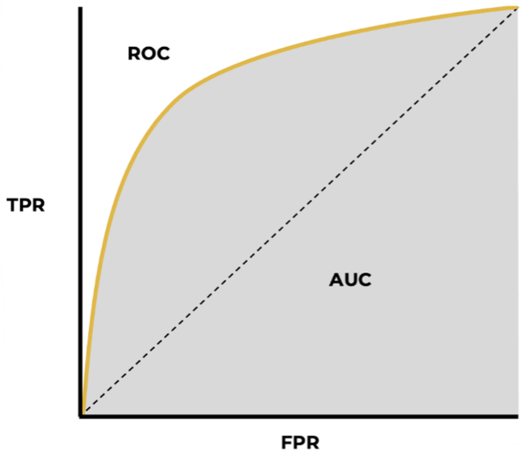
Mean Squared Error (MSE) – measures the average squared difference between the estimated values and the actual value. when our task is to evaluate continuous numbers rather than discrete categories.

Area Under the Curve (AUC) Analysis – The objective measure of the ability of a classifier to distinguish between binary classes, such as predicting the presence or absence of injury. It is used as a summary of the Receiver Operator Characteristic (ROC; see below) curve.

The higher the AUC, the better the performance of the model at distinguishing between the positive and negative classes.

- When AUC = 1, the classifier is able to perfectly distinguish between all the Positive and the Negative class points correctly.
- When AUC is between 0.5 and 1, there is a slightly better than random to high chance that the classifier will be able to distinguish the positive class values from the negative class values.
- When AUC = 0.5, then the classifier is not able to distinguish between Positive and Negative class points.

Receiver Operating Characteristic Curve (ROC) – is an evaluation metric for binary classification problems. It is a probability curve that plots the True Positive Rate (Sensitivity) against the False Positive Rate (Specificity) at various threshold values. A higher X-axis value indicates a higher number of False Positives than True Negatives. While a higher Y-axis value indicates a higher number of True Positives than False Negatives.



Terminology:

ROC - Receiver Operating Characteristic
AUC - Area Under the Curve
TPR - True Positive Rate
FPR - False Positive Rate

F1-score – combines the precision and recall of a classifier into a single metric by taking their harmonic mean. It is primarily used to compare the performance of two classifiers. The F1-score is dependent on the selected classification threshold as part of the tuning process.

Inputs

Data quantity and quality

Inputs: *Data quantity and quality*

Jo Clubb
Global Performance Insights

When surveying the industry on AI in sport, a number of queries focussed on the system inputs. For instance, practitioners asked:

Does elite sport currently have sufficient data available to train models? And how do we know if our data inputs are good enough?

These are important questions. However, as we will discuss, the term "enough" is perhaps reducing the discussion to a 'yes' or 'no' when in reality, it is not dichotomous but a more nuanced consideration. Therefore, the question should perhaps be positioned as follows;

What is the minimum viable dataset in terms of size and quality to train models in elite sports?

Are we dealing with 'Big Data'?

A little over a decade ago, it was a novelty to have tracking technology used on the training field in professional sports. Now, athletes are often tracked every time they take the field, as well as in the weight room, during rehabilitation, and even in the swimming pool.

Many major leagues not only now permit in-game tracking, but use a league-wide provider that enables access to all games for all teams. As a result, there is increasing integration between physical and technical/tactical data. Within teams, tracking data is frequently supplemented with jump testing, subjective measures, strength tests, body composition assessments, and many more.

This seems like "Big Data". *But is it?* Perhaps surprisingly (*to me at least*), Big Data is not defined through quantification but by the characteristics of the data. Specifically, by the three V's: **Variety**, **Volume**, **Velocity**. Put simply, Oracle describes big data as "larger, more complex data sets, especially from new data sources".

It is worth noting that some argue **Value** and **Veracity** should also be added to the V's to better define Big Data. *More on Veracity later...*

Tracking data and in particular, in-game data, appears to meet the 'Three-V tipping point' for Big Data; the point at which traditional data management and analysis becomes inadequate (Gandomi and Haider, 2015)[1]. So, it is not surprising to witness collaboration between sports and computer science to research tactical behaviour, such as the framework by Goes and colleagues (2021)[2] for tactical performance analysis.

1 https://doi.org/10.1016/j.ijinfomgt.2014.10.007
2 https://doi.org/10.1080/17461391.2020.1747552

Inputs: *Data quantity and quality*

Tal Brown
Zone7 Co-Founder & CEO

What is Zone7's approach to using "Big Data" to model injury risk and load management?

The first step is to define the initial data assets that are needed to model a problem, in this case injury risk forecasting and workload load management.

This is a multi-dimensional problem so ideally we'd like to ingest descriptive data that covers workload and injury data from multiple teams and leagues. This typically touches on tracking technologies, game videos, strength assessments etc. The larger the data lake, the better (as long as it's uniformly "clean").

Within football, the reality is that not all leagues and teams use the same sensors. For example, teams often employ Global Positioning System (GPS) and accelerometry-generated data during training, but camera-based optical tracking technologies during competitive matches. Yet, these between-system differences mean that such data is not easily interchangeable (Taberner et al., 2019)[1] and special care is needed here, otherwise the inputs are corrupted.

A common approach for this is to rely on an Extract, Transform, Load (ETL) architecture. ETL acts as the scaffolding through which the data can be normalised. This process ensures disparate datasets are standardised and therefore, can be used harmoniously on any longitudinal injury risk analysis.

Riding the 'Data Quantity Wave'

In applied practice, we can ride peaks and troughs in data quantity. External load data can be abundant, thanks largely to the in-game tracking already mentioned, as well as the less invasive nature of certain devices (e.g., optical tracking or wearables integrated into equipment/clothing).

Meanwhile, internal load is sometimes neglected, perhaps due to the increased cooperation required, despite the added value it could provide. Often, subjective measures of internal load (namely, Rating of Perceived Exertion) and/or response (e.g., self-report questionnaires) are used, although these rely on the pillars of athlete self-consciousness, autonomy, and honesty (Montull et al., 2022)[2].

During pre-season screening, we may be overwhelmed by data. This generally recedes to varying levels as the season gets underway. Personally, I think a one-off assessment can be used to guide individualised programmes in an effective manner.

That said, clearly there can be benefits to increased data collection frequency, with the caveat being that the data is then utilised.

1 https://doi.org/10.1080/24733938.2019.1634279
2 https://doi.org/10.1186/s40798-022-00432-z

Inputs: *Data quantity and quality*

I would love an objective way to quantify the greater value associated with the greater burden of repeat testing. “Training as Testing” methodologies enable more consistent data collection.

These include fitness-fatigue assessments during standardised drills, integrating regular jump testing to track neuromuscular fatigue, velocity-based training data collection, and measures of eccentric and/or isometric strength incorporated within a gym training programme.

Theoretically, this increased quantity pertaining to an athlete’s fitness-fatigue relationship and physical capacities should augment a practitioner’s decision making relating to health and performance. Is this the case? And if so, can AI enhance this process?

Can multiple input categories be used effectively in a model? Specifically, can external workload data combine with other datasets like strength assessments, fatigue, RPE, etc?

> **“Building a culture that respects and values clean data practices starts with us.”**

Tal Brown
Zone7 Co-Founder & CEO

The Zone7 philosophy is that practitioners and organisations know best and they determine which datasets should be included in the analysis. That said, a go-no-go criteria is helpful to assess whether a specific dataset deemed relevant is “usable” in a predictive model aimed to address load management and risk forecasting:

***Cadence -** daily outputs require daily (or weekly) inputs. Hence this appears to be a strong argument for synergising “testing as training”. So, while yearly strength assessments are hugely valuable to practitioners, they may be difficult for inclusion in a model aiming to produce daily results.*

***Volume -** a certain critical mass of data is needed to demonstrate validity of the model and a baseline per athlete.*

Hence, some datasets may require an initial ‘ramp up’ period of collecting them before they can be included in a predictive model. That said, they may very well be useful to practitioners from day-one of their inclusion for “human analysis” methods.

***Quality -** The other key parameter to consider is data quality, which we expand on below.*

Inputs: *Data quantity and quality*

Jo Clubb
Global Performance Insights

Taking pride in data quality

Of course, data quantity is not the only factor to consider with inputs. It is no good collecting a mountain of data if it is unusable. Garbage In Garbage Out (GIGO) is a data science concept that contends poor quality inputs will result in faulty outputs. Despite the common sense of this concept, IBM estimated the cost of poor data quality to the U.S. economy at over $3 trillion annually.

Data quality should always be a priority to sports science practitioners. We are "Data Stewards", responsible for ensuring sufficient quality throughout data collection, storage, and analysis.

Coaches, management, and the athletes themselves should be able to trust the information presented to them. If data is to be used for decision making, the outputs will only be as good as the inputs. Building a culture that respects and values clean data practices starts with us.

Data veracity has multiple definitions but is generally considered as the accuracy and fidelity of data (Reimer and Madigan, 2018)[1]. A systematic review conducted veracity analysis on load monitoring publications in professional soccer found 73% of the studies did not report veracity metrics, such as coefficient of variation (CV), intraclass correlation coefficient (ICC), and the standard error of measurement (SEM) (Claudino et al., 2021)[2].

We may consider such statistical measures to be innately linked to a particular technology. Yet, often our own processes directly affect data quantity. We previously published a 'Clean Tracking Data Checklist' to promote a systematic process in external load data collection (Torres-Ronda et al., 2021)[3].

In another example, ensuring accurate body weight of a subject on a force plate, via a stationary weighing phase is important to maintain data quality within and between trials and athletes (McMahon et al., 2018)[4].

There are also factors that while not directly within our control, sit within our sphere of influence. For instance, earlier I mentioned how self-reported, subjective measures rely on athlete honesty (Montull et al., 2022)[5]. Athlete buy-in can be supported through education and communication.

Explaining the purpose and use of such data is key to trying to gain their trust in the process. Similarly, coach support has been shown to influence compliance of self-reported measures (Saw et al., 2015)[6].

Clean and repeated data collection and management practices do not seem exciting, but they lay the foundation for effective data analysis, whether that is by human or machine.

1 https://doi.org/10.1177/1460458217744369
2 https://doi.org/10.3390/app11146479
3 sportperfsci.com/wp-content/uploads/2022/03/SPSR159_Torres-Ronda
4 http://dx.doi.org/10.1519/SSC.0000000000000375
5 https://doi.org/10.1186/s40798-022-00432-z
6 https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4657415/

Inputs: *Data quantity and quality*

Eyal Eliakim
Zone7 Co-Founder & CTO

What are the best practices on regulating the quality of data that is ingested? *Do Zone7 provide insight to practitioners around their specific data quality and/or cleaning needs?*

Datasets need to be validated for quality to ensure any outputs generated are contextual, meaningful and potentially impactful. The GIGO maxim is at the forefront of any data science quality control process.

To be effective in a predictive model, it is vital that practitioners use consistent data collection and cleaning practices as described above. Processes like checking for velocity spikes, being consistent with thresholds/bands, and managing the athletes in the correct sessions and drills.

It is not uncommon for Zone7's automated "data diagnostics" to uncover glitches that have a negative impact on the data integrity in a specific environment. We have the tools to assist practitioners and their organisation to ensure data quality is sufficient. Measures are in place to ensure each new contributing dataset is not only an appropriately valid dataset, but also one where ad-hoc anomalies are identified and removed so that they do not pollute the data lake.

Image: Zone7 Performance Director Rich Buchanan during his tenure at AFC Swansea

Inputs: *Data quantity and quality*

Jo Clubb
Global Performance Insights

Considering our blind spots

We should also consider potential influential factors we are not currently capturing. These may be 'known-unknowns', exemplified perhaps by the gut microbiome and potential links with injury risk. Further, there are clearly things we do not yet know about human physiology, performance and injury risk: the 'unknown-unknowns'.

As Nate Silver wrote in 'The Signal and The Noise';

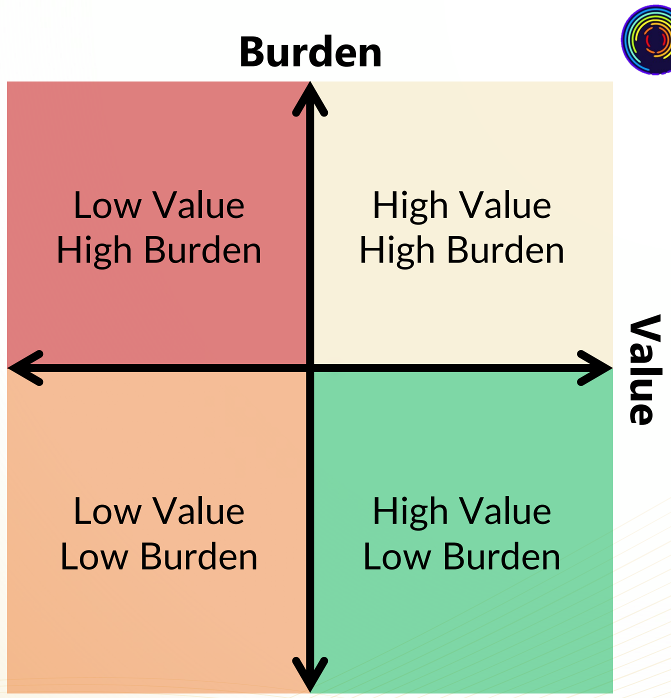
> **One of the most pernicious [biases] is to assume that if something cannot easily be quantified, it does not matter.**

Another potential limitation to our inputs is the instantaneous nature of our testing and, to some degree, monitoring assessments.

Even a daily cadence of data collection does not reflect how physiology changes throughout a single day. Injury risk is dynamic. In a utopian sports science world, we would collect such measures in a continuous manner and subsequently, have access to real-time injury risk. Though that would open a whole other can of worms...

That said, I think we have a responsibility to induce more value out of the data we already collect. While some argue for a 24-hour approach to athlete monitoring (Sperlich and Holmberg, 2017)[1], others decree that it has already gone too far (Powles and Walsh, 2022)[2].

My Value-Burden Matrix has been well received, I believe, because there is consensus that we owe it to our athletes to optimise this relationship; maximising value from the data collection burden that we place on them.



1 https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5557737/
2 https://theconversation.com/has-the-monitoring-of-professional-athletes-intimate-information-gone-too-far-180890

Value Burden Matrix. For more see:
https://www.globalperformanceinsights.com/post/juggling-value-burden-in-data-collection

Inputs: *Data quantity and quality*

Tal Brown
Zone7 Co-Founder & CEO

How can we know or assess if AI is adding greater value to our processes? What defines success?

Whether AI is effective is ultimately a subjective statement, driven by the collision between the model's mathematical accuracy and the specific needs of the environment.

The mathematical "success" of a predictive model is driven by the combination of the dataset size and the quality of the code/algorithms. An algorithm looking for patterns associated with (for example) soccer-related hamstring injuries in women's soccer will get better the more "examples" of data/injuries it can "learn" from. By "better" we typically mean less mistakes (either less false positives, or less false negatives or both).

Some algorithms will detect 90% of events with lots of false alarms, and others will detect 65% of events with very few false alarms. There's always a trade-off. On the other hand, usability is subjective. Some environments might be much less tolerable of false positives than others. In sports, a large staff of physios and sport scientists might be better equipped to deal with a large number of 'alerts' per day/ week.

Hence, their accuracy needs are different. This can also change per group of athletes (e.g., by position) or seasonality.

So our suggestion when answering "do we have enough data" is to look at several factors in tandem:

1. *The dataset available in-house in terms of size and quality, and*
2. *The dataset available to train the algorithms with, and*
3. *The algorithm's usability in the specific context it is being deployed in.*

Inputs: *Data quantity and quality*

Jo Clubb
Global Performance Insights

Improving value in the data we already collect

Whilst sports science datasets are not as big as those collected in some other fields, we are still *trying* to manage large, complex data sets coming from new sources, that require a high velocity to analyse and disseminate.

Whether we turn to man or machine to delve deeper, the human practitioner has an obligation to optimise the quantity and quality of the data collected from their athletes. Consider the following as good practices when it comes to this:

1. Employ a regular data collection cadence
2. Consider the Value-Burden Matrix when introducing a new data collection process
3. Maintain data stability and document any changes to settings, in particular bands/thresholds with tracking technology
4. Use systematic data collection processes and compile these as protocols to ensure consistency across staff and seasons
5. Regularly conduct tests for data integrity (Zone7 can help with this).

Personally, my exploration of AI thus far has highlighted that the discussion is more nuanced than I expected. There is no magic number, no binary threshold of "enough", when it comes to quantity and quality. Clearly though, there is an influence on the quality of outputs. Do they determine success?

> ***... like any type of analysis, it comes down to how it is contextualised, interpreted, and actioned...***

To some degree yes, but like any type of analysis, it comes down to how it is contextualised, interpreted, and actioned within each specific environment. These will be topics that we will explore in future articles, starting with context.

Context

Incorporating situational factors

Context: *Incorporating situational factors*

Jo Clubb
Global Performance Insights

So far in our deep dive into AI and injury risk and performance modelling, we have defined important terminology and explored data quantity and quality.

As we move through the data pipeline to analysis, there was one term that kept coming up: Context. Practitioners ask:

"We always talk about the context of data. How does AI take account of context in the data?"

So why is context at the forefront of practitioner minds? And, can AI include relevant context to make it more usable? In this post, we'll explore these questions.

Why Context is Key

Implementing sports science should always incorporate a critical appraisal of the context. Textbooks present theories in an unequivocal manner, yet the application of scientific theory requires a wider appreciation of the circumstance.

Performance and injury are complex constructs. Contextual factors influence the web of determinants that drive their emergent outcomes (Bittencourt et al., 2016)[1].

Some constraints, like the rules of the game, are (mostly) fixed. Meanwhile, there are a multitude of other factors, such as schedules, player availability, and fitness-fatigue relationships, that are in continuous flux.

The sport's demands, playing positions, and the individual athlete are each contextual foundations.

Additionally, we are required to vary our perspective to the changing contextual landscape, particularly across different times of the season. Let's illustrate these factors with examples.

Context starts with the sport and positional demands

I've previously discussed the challenge of updating my own practitioner lens to transition from football to ice hockey, and then later to American football. Training theory remains the same, but the context within which it is applied is grossly different.

Ice hockey is played on ice (obviously!)... But how does this influence load monitoring? What does the difference in ground reaction forces (ice v land) mean for external load tracking?

How do we account for gliding, whereby athletes are moving over the ice without physiological cost to that movement?

Meanwhile, American football presents the most diverse, within-squad load measurement and management challenge of any team sport. The positional demands greatly differ. Consider, for example, the varied physical requirements of a quarterback, a lineman, a wide receiver, and a kicker, to name a few positions.

1 http://dx.doi.org/10.1136/bjsports-2015-095850

Context: *Incorporating situational factors*

In addition, "Special Teams" demands can be wildly different compared to a player's primary role on Offense/Defense.

We can undoubtedly learn much from other sports. Yet, the unique context of each requires translation of processes, rather than transplanting them. As we illustrated in our Sports Medicine Open review of tracking systems in team sports, sport- and position-specific analyses are required to provide meaningful insights into athlete management.

Eyal Eliakim
Zone7 Co-Founder & CTO

How can AI take into account the contextual demands across the different sports and positions?

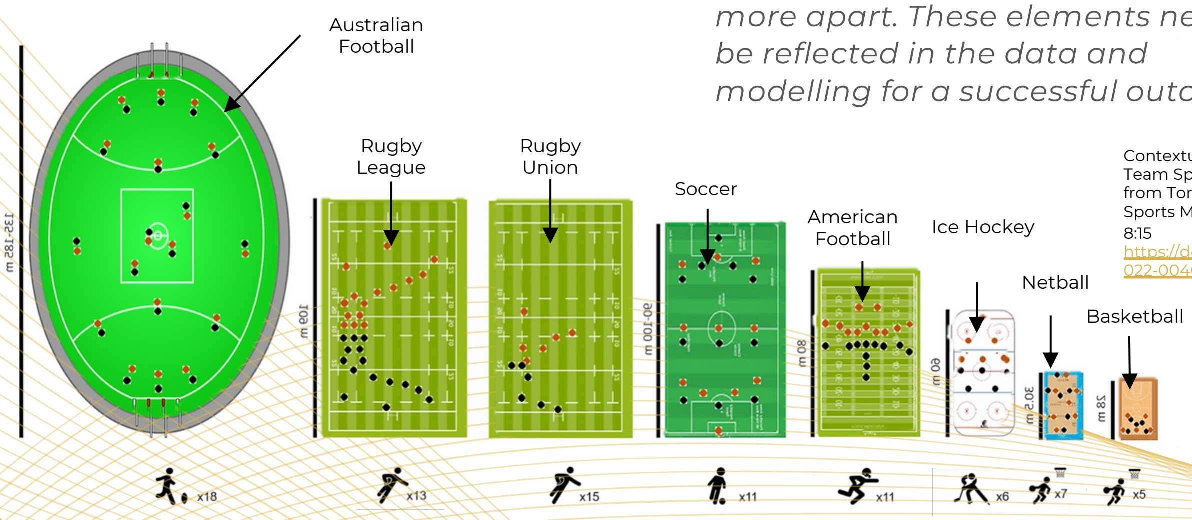
For a data science approach to make sense in sports performance, it must 'speak' the language of the specific sport. This is all about injecting the "context" discussed above into the predictive models.

Here are some questions Zone7 seek to answer when diving into a new sport/ environment

1. ***Injury epidemiology*** *per sport varies dramatically and is influenced in different ways by the contact and physical demands of the sport. What are common injuries and mechanisms in this sport? Do they interact in subtle ways?*

2. ***Micro seasonality*** *- as an example, NBA, like NHL and MLB, have shorter micro-cycles than the NFL because they compete more frequently. On the other extreme, there are endurance athletes that compete once every several weeks or months. It is therefore important to ask ourselves; what are common periodisation practices in this sport and how do they interact with match/ competition cadence?*

3. ***Macro seasonality*** *- while soccer is played for 8 months with a short preseason, American Football environments have dramatically different season durations as well as pre-season characteristics. Tennis and long distance running are even more apart. These elements need to be reflected in the data and modelling for a successful outcome.*



Contextual Factors Across Team Sports. Figure adapted from Torres-Ronda et al. Sports Medicine - Open (2022) 8:15
https://doi.org/10.1186/s40798-022-00408-z

Context: *Incorporating situational factors*

Jo Clubb
Global Performance Insights

An international tournament like no other?

Dynamic contextual factors require constant consideration for athlete management. It can be the difference between winning and losing so-called "unwinnable games". These factors include the competition calendar, opposition, game times, locations, environmental factors, and team availability. Such contextual fluidity can present a test of our translation abilities. A notable illustration of this is the recent 2022 FIFA Men's World Cup.

For the Member Associations, Qatar presented the usual challenges of an international tournament: travel fatigue and jet lag, where and how to set up their base, training and recovery management during fixture congestion, to name only a few.

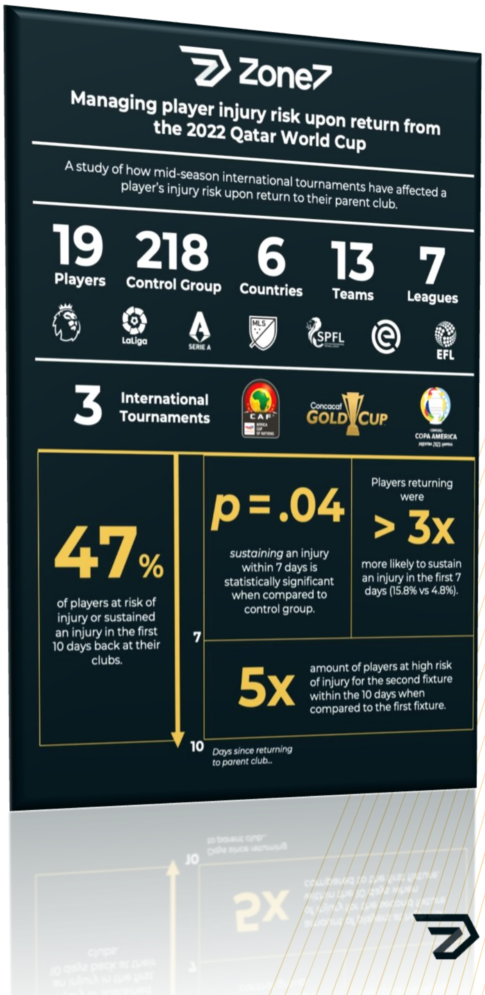
Yet, the most noteworthy context is the competition timing. Holding the World Cup in the middle of the (European) season for the first time posed a unique challenge to domestic teams.

They face continuously shifting contextual factors within their playing squad; in injury availability, training and game loads, fitness-fatigue status, and travel demands. Not to mention the psychological impact of varying performance and success at the tournament.

All factors outside of their control in the international playing group for the time each country remains in the tournament. Additionally, they have to manage training and recovery factors within the playing group that remain with the domestic team.

Thereafter, they have to blend the training and management needs of each athlete, as teams retake the field for the second half of the domestic season.

Ahead of the tournament, this was explored in more depth by Zone7's Data Research Analyst Ben Mackenzie using data from other mid-season tournaments.



Context: *Incorporating situational factors*

Rich Buchanan
Zone7 Performance Director

How can AI take into context the changing competition schedule and team-specific approaches to periodisation and recovery?

The success of AI to account for context depends largely on the strategy and inputs from the human operator to aid with contextual appreciation. Practitioners are able to input the team's competitive schedule, along with their specific periodisation model to the system, for instance. Clearly periodisation context is key to any load management decision.

An example in relation to the aforementioned Qatari World Cup tournament would be for the practitioners at the domestic clubs to provide the system with information around the players that remain under their care.

Factors that could be provided to the AI system include time off, friendly game schedules, altered training micro-cycles, and potential travel demands.

All this before potentially adding in any relevant information they may be able to acquire from the international teams about their players competing at the World Cup tournament.

Does Zone7's AI take into context the individual athlete?

Contextualising for the individual athlete is an integral part of how Zone7 model the reality of athletes operating in a highly demanding environment.

Without diving too deeply into Zone7's intellectual property, they create a holistic athlete profile that connects the dots across as many data points as the environment can provide, including:

- *Match/competition and training external workload data*
- *Internal workload*
- *Strength assessments*
- *Other environmental and historical*
- *Injury history*

This profile is in constant flux and changes every day or even in real-time sometimes. To "use" this profile in a forecasting model and compare one athlete to another, or apply pattern recognition algorithms across a large dataset, a normalisation process occurs. This is sometimes called 'baselining' and involves a mathematical analysis to extract an athlete's baseline, and measure how today's profile differs from the baseline.

Context: *Incorporating situational factors*

Jo Clubb
Global Performance Insights

Individual athletes are more than just data points

In our open-access review, led by Dr Stephen West, we illustrated a wide range of contextual factors that influence athlete injury risk and readiness to perform. A large portion of this figure considered athlete level factors.

Part of the beauty of sport is the diversity in its competitors. This emphasises the need to embrace individuality. Interestingly, one of the principles of individuality is context, reinforcing how context and individuality are interwoven.

Averages and group level analyses serve their purpose in sports science and analytics. But our analysis is clearly benefitted when context for individual athletes is taken into account.

One of the most valuable contextual factors for athlete management is injury history. It is widely accepted to be one of, if not, the strongest predictor of future injury. Workload management decisions are often different for an older athlete with a history of soft tissue injuries than other athletes, for example.

While prior injuries and age are thought of as non-modifiable factors, research has suggested that increasing risk associated with these factors can be mitigated by eccentric hamstring strength (Opar et al., 2015)[1]. As such, connecting sports science data streams, including potential moderators and mediators to workload, is vital to establishing a holistic athlete profile.

Context as a Double Edged Sword

Clearly, context is an essential ingredient for sports science analysis and interpretation.

Yet, I believe it is important to be continually reflective and critical of how we view context. Consider, for example, the following statements:

- "He is injury prone"
- "She is weak"
- "His body doesn't cope with high training loads"
- "They're a difficult opponent"

Are these representing context? Or bias? When does so-called context actually become bias?

Perhaps context is in danger of declining into bias, when it is a factor that is not, nor cannot be, backed by objective information. The challenge, therefore, is to support our contextual factors with objectivity.

Can we justify the context of "injury prone" with injury history? Can we objectively determine "weak" through strength diagnostics? Do we change our minds if the evidence suggests so?

1 https://doi.org/10.1249/MSS.0000000000000465

Context: *Incorporating situational factors*

Much like the need to consider the data input quality, we have a responsibility to be critical of the quality and suitability of the context we add to analytics and AI systems.

This responsibility is illustrated in the Netflix documentary, Coded Bias. One summary[1] of the documentary from PBS describes how:

> **… the very machine-learning algorithms intended to avoid prejudice are only as unbiased as the humans and historical data programming them.**

Thus, it is not the responsibility of the machine. We should be reflective of our own contextual factors, and be wary of when our so-called "context" actually meets bias.

1 www.pbs.org/independentlens/documentaries/coded-bias/

Tal Brown
Zone7 Co-Founder & CEO

How can a predictive model avoid bias and help practitioners stay away from blind spots?

Bias cannot be detected from within. Once a system is in motion, it is very difficult to detect bias, or drift in accuracy. So the healthy approach in data science is to accept that there will always be bias and to 'bake' as many preventive measures into every step of the process as possible.

It's important for any system, whether spreadsheet macros or machine learning code, to continuously evaluate accuracy against a solid baseline, much like elite practitioners who reflect, review and evolve their own professional practices. This might take time and effort, whilst also going against what we originally assumed to be true, but is the only way to deepen understanding and create confidence among those relying on these insights.

The Zone7 process of creating forecasting models includes the most rigorous cross validation methods and ongoing monitoring of results, both automatically and by expert operators. In addition, models need refreshing periodically, which again - is a high touch process that must be done with care and the utmost integrity.

In addition, it is key to provide the practitioner with a transparent 'snapshot' of the bias risk in the system. Only transparency can breed success and positive improvements.

Context: *Incorporating situational factors*

Final Thoughts

As demonstrated by the questions which I received, context is clearly at the forefront of practitioners' minds. This comes as no surprise. We are acutely aware of the need to consider how various factors, such as the sport, position, individual athlete, and time in season, shape our athlete management decisions. As such, this context must be also injected into any analytics and AI. Responsibility ultimately lies with the practitioner to incorporate appropriate contextual factors, while trying to be mindful of their personal biases.

> ***We're blind to our blindness. We have very little idea of how little we know. We're not designed to know how little we know.***

Image: Jo Clubb taking notes whilst working with the NFL's Buffalo Bills

Transparency

Validation of and access to the algorithms

Transparency *Validation of and access to the algorithms*

Jo Clubb
Global Performance Insights

When it comes to training load management, sports scientists are constantly trying to manage the tension between simplicity and complexity, and between usability and fidelity.

The acute: chronic workload ratio (ACWR) came under attack for taking an overly simplistic analysis approach to a complex problem. At the other end of the complexity spectrum, there is trepidation about “black box” structures; those which are opaque, non-intuitive, and difficult to interpret in relation to decision-making (Teng et al., 2022)[1].

Sports scientists have a desire, a need in fact, to understand the results that machines output. We strive to “unpack the black box” (Malone et al., 2017)[2], to understand the inner workings of force plates and tracking technologies. Our interventions and recommendations are built upon trust; trust that is itself built upon the ability to reason our decision making.

1 https://doi.org/10.1007/s00530-022-00960-4
2 https://doi.org/10.1123/ijspp.2016-0236

It was unsurprising then that transparency and black boxes came up as a key AI topic for sports science practitioners.

In this report so far, Zone7 has added to the discussion on the AI Dictionary for sports scientists, sports science data inputs, and the importance of contextual factors. As we move onto the analysis part of the data pipeline, they have kindly agreed to answer practitioner questions relating to analysis, transparency, and validation.

Image: Jo Clubb speaking at an event on behalf of the NFL’s Buffalo Bills

Transparency *Validation of and access to the algorithms*

Tal Brown
Zone7 Co-Founder & CEO

Firstly, can you give an overview of Zone7's approach to data analysis?

- *First principles - access to comprehensive datasets that touch on what we, and the practitioners think are key important inputs. Data should be in digital form and collected through validated technologies.*
- *Be tech-agnostic and reliably transform workload, personal history and other inputs from multiple trusted sources into a robust multi-team data lake that reflects data quality as well as volume.*
- *Have a quantified sense of how consistent and clean different datasets are, so that corrupt or sparse datasets can be identified and bypassed, as discussed earlier in this report.*
- *Careful care is given to the completeness of the timeline. In other words, identify and have clear criteria on how to deal with data gaps in the timeline. Some examples are soccer/football players that leave for international fixtures, or temporary lack of data due to malfunctions or decisions by practitioners.*
- *Reflect the environmental needs of both the organisation's and individual user's attitudes towards risk. This is manifested in digital data articulating the relevant context, as well as calibration to conform to the environment's preferences.*
- *Perform data analysis efficiently with an understanding that the human performance space is multifactorial.*
- *Dynamically adapt and learn over time to improve injury risk forecast accuracy as variables change and more data is collected, meaning the system's sensitivity and specificity are expected to improve.*
- *Appropriately translate data trends and clearly communicate actionable insights for the human end-users to apply as deemed appropriate.*

Does Zone7 operate with a 'black box' approach?

It is important that the underpinnings of machine learning (ML)/ deep learning (DL) solutions deployed in a medical and sport science environment be as transparent as possible, in terms of how the algorithm models have been developed and validated and by whom.

Interpretable black boxes describe big data being ingested, transformed and analysed by ML/DL processes, resulting in insights which humans can interpret and act upon as deemed appropriate.

When systems are created "in-house", it is typically easier to access these insights and the reasoning behind them. However, commercial data science solutions typically do not provide their source code and the human operator might be left with bottom line insights that are difficult to reason through.

To avoid these pitfalls, Zone7 adheres to two key principles:

1. *The validation process should be as transparent and strict as any in-house solution.*
2. *While source code may not be provided, explainability components are an absolute must to provide operators with the ability to "reason through" insights before implementing.*

Transparency *Validation of and access to the algorithms*

Rich Buchanan
Zone7 Performance Director

Where have you seen black box projects fail?

There are two underlying dangers of working 'blindly' with a black box system:

1. *Insufficient help or context is provided to understand the reasoning driving the risk assessments, hence the practitioner must either spend considerable time 'reverse engineering' or go on blind faith.*
2. *For the system to overstep and encourage insights to be interpreted as causality related. Insights must be clearly tagged with such context and robust validation must be provided if causality is suggested.*

Correlation is not necessarily an explanation of causality. Yet, it is important to acknowledge that ingested datasets should be representative of what is trying to be interpreted as possible causality by the algorithms. Such an example in the sporting context is a GPS and accelerometery dataset, which represent an insight into an athlete's mechanical and locomotive output when participating in their specific sporting activities.

A common example (rightly or wrongly) is the broadly accepted notion that external workload datasets from player tracking technologies indicate some level of causality relationship towards amplifying performance potential and understanding injury risk.

Presenting the algorithms with datasets that have no known causality or relationship to the output would be unwise. It is therefore important that domain experts apply their professional knowledge when deciding what datasets are important for algorithms to examine, in order to seek a deeper understanding of potential causal relationships. It is also not uncommon for datasets collected by commercially available solutions (e.g., strength assessments, GPS) to be ingested along with proprietary/in-house datasets (e.g., strength and biomechanical assessments).

> ***Zone7 avoids overfitting by adhering to strict operational principles of absolute separation between training, validation data, and test data.***

Machine learning techniques are said to be at risk of overfitting – how does Zone7 reduce this risk?

Overfitting refers to cases where an algorithm's results are skewed or biased to show 'better' results. This is typically associated with a (usually unintentional) leakage of data from the dataset used to test the algorithm and generate results, and the dataset used to train the algorithm.

Zone7 avoids overfitting by adhering to strict operational principles of absolute separation between training, validation data, and test data. When creating validation studies per client, this principle is manifested by never including the client's data in the training dataset.

Transparency *Validation of and access to the algorithms*

Tal Brown
Zone7 Co-Founder & CEO

How does Zone7 balance the need to keep the company's Intellectual Property (IP) private with the desire to be transparent?

Transparency has multiple levels, and each organisation or practitioner may require a different level depending on their experience and/ or needs. Zone7 operates with the following principles at the core of our engagements with clients:

1. *Detailed validation studies are shared openly with clients. These provide an opportunity to learn how the algorithm is tested against other datasets.*

2. *Custom validation studies may be created for clients, providing them with full control over inputs and outputs. Results from these projects include all the raw output data so clients can understand how key metrics are calculated.*

3. *Transparency is also about explainability of the results - all our products include an explainability layer which breakdown the "why" - What specific parameters are driving risk at any moment? These are both visualised and provided as data for clients to engage and investigate.*

4. *Explainability can also be generated by simulating modifications of controllable variables and asking a model "what if" questions. This is also a key part in building trust and is shared openly with the environment operators.*

5. For clients with in-house data science teams, we also provide documentation and detailed workshops about the internal workings of the platform. That said, Zone7 is not an open source platform, hence source code is not provided.

One of the claims frequently made against ML is that it is not actionable. How does Zone7 turn algorithms into interventions?

The holy grail of data science is to drive impact through action. To make this happen, any predictive technology must provide the following:

1. *Agree on the desired metrics both for retrospective analysis and for live impact on the environment (e.g., how we measure player availability and changes to it)*

2. *Accurate and trusted forecasting*

3. *Contextual explainability*

4. *Provide the operator with potential modifications to controllable variables that will hopefully drive towards the desired impact.*

Zone7 provides all of these components to ensure expert practitioners are empowered with a time efficient and deeper insight into what the data they are collecting is indicating. Insights include detailed risk forecasting profiles, reasons behind each forecast as well as load management scenarios that are 'human readable' and can be easily interpreted.

Transparency *Validation of and access to the algorithms*

Gavin Benjafield, Assistant Coach & Performance Director of Los Angeles FC, recently shared how he actions the Zone7 insights day-to-day at LAFC in a webinar that is available to watch on our website.

How does your analysis support your clients, but also potentially the wider sports science community, in understanding injury risk forecasting and training load management?

Through interrogation of the data lake, we are able to explore and disseminate insights to specific clients, as well as the wider community. We frequently engage with clients and industry experts to learn what the industry needs and is curious about. That's how this report came about; starting with the questions and comments that came from the sports science community.

By providing answers to these questions, significant or not, we feel we are able to contribute to the sports science landscape. As scientists, we are driven to share our findings to help improve knowledge and further understanding.

We're trying to focus on those questions most relevant, particularly at time-critical moments. We explored pre-season injury statistics prior to this season's pre-season, then invited one of our industry experts, Damian Roden, to provide his personal perspective on the findings. Recently we explored injury risk for players involved in this unique men's World Cup, by analysing injury trends from other mid-season tournaments.

> ***"We frequently engage with clients and industry experts to learn what the industry needs and is curious about."***

Image: LAFC's Gavin Benjafield consults a player during a training session.

Outputs

Injury risk and load management as outcomes

Outputs: *Injury risk and load management as outcomes*

Jo Clubb
Global Performance Insights

We have previously defined notable terms in our AI Dictionary, considered the importance of data quantity and quality as inputs and injecting contextual factors, and most recently, discussed analytics transparency with Zone7.

In this next post on AI in sports science, we turn our attention to the system outputs.

Specifically, we are going to look at the delivery portion of the data pipeline. However, before discussing how practitioners can integrate the system into their daily workflow, it is worth becoming more familiar with the information the user receives.

When it comes to system output design, there are a multitude of decisions that need to be made with great care and intention by the provider.

Why are the outputs important?

AI outputs have the potential to bamboozle human operators. IBM's AI-driven Deep Blue software infamously beat world chess champion Gary Kasparov in 1997. One move was particularly unusual and mentally threw Kasparov off his game; he could not comprehend why the system would output this move. In reality, it had chosen a random move!

"One of Deep Blue's designers has said that when a glitch prevented the computer from selecting one of the moves it had analysed, it instead made a random move that Kasparov misinterpreted as a deeper strategy…

"The world champion was supposedly so shaken by what he saw as the machine's superior intelligence that he was unable to recover his composure and played too cautiously from then on… He over-thought some of the machine's moves and became unnecessarily anxious about its abilities, making errors that ultimately led to his defeat." The Conversation[1]

Of course, in this example the AI and the operator were competitors. In the context of injury risk analysis and load management design, the operator and "system" are teammates. Yet, this story illustrates the potential negative, even unsettling impact outputs can have when they are not understood.

1 https://theconversation.com/twenty-years-on-from-deep-blue-vs-kasparov-how-a-chess-match-started-the-big-data-revolution-76882

Outputs: *Injury risk and load management as outcomes*

As such, any team or company creating AI systems has a great responsibility when designing these outputs. Model results need to be transformed into outputs that provide the right level of detail, an appropriate rationalisation and are actionable. The choices behind this transformation must be difficult. Hence, it is hugely important for providers to engage with experts in the applied environment and use their domain knowledge when developing output design.

As is often the case in data science, it presents a delicate balance between usability and information overload and finding the right trade-off between reductionism and oversimplification.

Injury risk as the output

Injury risk forecasting has arguably become one of the most prevalent uses of AI in sports science currently. It is worth noting the emphasis on injury risk, rather than injury per se. It is not predicting injury occurrences in terms of their exact timing, severity or mechanism, but forecasting risk for a predefined time frame.

We are required to think probabilistically, much like we do with the weather forecast!

When I canvassed industry questions on AI, many practitioners wanted to know why the focus was on injury risk, rather than other variables, most notably, performance. There was concern that we may be limiting resilience or even performance itself, if our decision making is overly focussed on injury risk. This is an important discussion, given the negative connotation around sports science promoting a risk-averse approach to performance.

I had this same concern but as I've become more familiar with AI in sports science, I understand the reasoning. Data science solutions benefit from an objective outcome. Injuries are (mostly) binary events that are documented objectively and consistently. Yes, in actuality, they exist on a continuum and there are many grey areas but, we do strive to objectively define and quantify them.

Outputs: *Injury risk and load management as outcomes*

As Eyal Eliakim, Co-founder and CTO at Zone7, told me;

> **One of the key things in building strong performing machine learning algorithms is clearly defining the event or trend we want the algorithms to learn. That is one of the reasons we feel injury risk forecasting is a very powerful use case - an injury is a very well-defined event.**

Meanwhile performance, particularly in the team sport setting, is complex. It is not purely about who runs the furthest or fastest. Therefore, performance is not easily defined universally, and game outcome may not even reflect performance!

Performance can differ based on positions, players, and opponents to name only a few contextual factors, making it difficult to model across a large set of leagues, teams, games, and players.

And yet, perhaps we're not best thinking about performance and injury risk as two opposite ends of the spectrum. Availability is correlated with performance and competition outcomes across many sports. It simply makes sense that having your players available for more games gives your team a better chance of winning.

Yes, some could take the approach too far and wrap players in cotton wool. But generally practitioners seem to be united in the need to maximise availability while promoting physical capacities and load exposure in the quest for high performance.

I'm interested in tools such as Zone7's micro-cycle simulator, for instance, that enable future workload projection up to seven days ahead of time. Injury risk may provide the foundation for this system, but it can be applied and interpreted with maximising load and performance in mind. Ultimately, this comes down to how the human operator acts on the information provided.

Let's dive deeper into the specifics of the outputs in relation to Zone7. *What do they look like and why? And, what does that mean in practical terms?*

Outputs: *Injury risk and load management as outcomes*

Tal Brown
Zone7 Co-Founder & CEO

What is Zone7's approach to output design?

The most important thing Zone7 has to consider when presenting its AI-derived outputs is determining and fully appreciating the needs of end users within the context of their specific environment. This is done to ensure the end user can apply the insights as they deem appropriate, thereby making our outputs more impactful.

Through translating mathematical models numbers, percentages and probabilities, Zone7 provides the human end user with information that would otherwise be unobtainable with such efficiency. Translation of our model results into our outputs is designed to make complex deep learning processes interpretable, digestible and ultimately actionable. Initially, this centres on converting numbers into more meaningful states, such as bucketing athletes into broad risk categories like Low, Medium or High Risk.

Further outputs are added to increase usability and trust for the practitioner applying Zone7's AI system:

Risk Factors
Specific parameters are highlighted as the greatest contributing risk factors to the injury risk forecast being presented. This important output allows practitioners to understand "Why" an athlete is identified as being at risk (e.g., is risk driven by overload patterns in X or underloading patterns in Y).

Risk Management
Other usable outputs provided to Zone7 clients are suggestions on potential workload modifications using historical reference training sessions, which if applied can help to mitigate the identified injury risk.

This information is presented as optimal value ranges for the parameters contributing to the risk alert, such as X amount of absolute sprinting distance or Y amount of high intensity decelerations.

Risk Simulation
As mentioned earlier, we also provide injury risk forecast simulations based on future projected workloads and parameters, as predetermined by practitioners. Whilst this is primarily intended to function with injury risk in mind, outputs from this tool could be utilised for optimising performance potential.

Discretising continuous data for risk forecasting purposes, is of course not without limitations, such as the dangers in context-loss driven by broad calibration thresholds. To determine suitable thresholds, the calibration process is based on each specific environment, conducted in a collaborative manner with practitioners, is a transparent process to foster trust and should be flexible to allow different 'configurations' to be implemented effectively.

As Ben Mackenzie, a Data Research Analyst at Zone7, recently discussed with Leaders in Performance, giving each team the power to calibrate the outputs according to their philosophy is important.

Outputs: *Injury risk and load management as outcomes*

> ***For each client, Zone7 can set individual thresholds for risk alerts which can be increased accordingly as individual management in a team setting is important.***
>
> ***Some environments desire a greater focus on physical development, while other departments want maximum availability for players and will therefore put less emphasis on that physical development to ensure availability. That is something Zone7 is able to cater for through calibrating attitude to risk – much like when making a financial investment.***

What Next?

As with any dashboard, the outputs should be chosen in a very deliberate manner. In sports science, we strive to reduce data abundance into simplistic, interpretable numbers; yet, reduce too much and essential detail may be lost.

When it comes to AI this responsibility is arguably greater, as data results and presentation guides the human operator that is consuming them. It is important that practitioners challenge providers on the 'hows' and 'whys' behind their output design.

Now that we have considered the outputs, we will move onto what us practitioners can do with the information - the application of the data within real-life settings.

Image: Zone7 CEO Tal Brown

Application

How do practitioners apply the outputs?

Application: *How do practitioners apply the outputs?*

Jo Clubb
Global Performance Insights

Man vs Machine: a debate fuelled by science fiction and Hollywood. As technology advances, the battle has garnered more and more attention...

In the red corner, Machine, armed with gigantic processing power that provides the ability to rapidly identify patterns or trends that may not be immediately apparent to humans. It does not tire, nor make silly mistakes purely due to "human error".

Yet, the machine is reliant on their human operators. It is hampered by Garbage In Garbage Out and relies on the human operator to input suitable contextual factors.

In the blue corner, Man, armed with expert domain knowledge. Sentient beings equipped with the day-to-day experience managing athletes in reality.

Man benefits from cognitive biases that help make sense of the world, providing mental shortcuts to reduce cognitive processes. Yet, these shortcuts can be a double-edged sword. For example, the 'affect heuristic' enables us to rely on our emotions, but can sometimes hamper our decisions.

The us vs the machine storyline is of course appealing to fiction writers. But these concerns are somewhat shared throughout the sports science community.

During my market research for this report, I received comments such as...

1. "My main concern is that some practitioners might overestimate the intelligence of that AI and not question the data/interpretations. 'Machine says so, it is very expensive/complex, so it should be truth!'"

2. "Nothing will outsmart human interaction, personal relationships & the human brain."

Can we bring the machine into our corner and battle together against injury and underperformance?! Can we utilise its strengths, while being mindful of its limitations? Maybe a peace process is required. If so, the notion of how the human operator can appropriately consume and implement such information from a machine teammate is a critical conversation.

Training Load Management Decision Making

When it comes to load management, we spend a lot of time discussing the collection and analysis processes: the technologies, metrics, timeframes, calculations (ACWR debate, anyone?!).

And yet, there appears to be less discussion about what we then DO with all that information.

So, before we even consider how AI might be integrated into that process, let's take a step back and consider the broader load management decision making processes.

Application: *How do practitioners apply the outputs?*

Contemporary training load philosophies build upon Banister's fitness-fatigue theory. A training stimulus results in two antagonistic responses; a short-term negative effect (fatigue) as well as a longer-term positive effect (fitness).

With repeated training exposures, an optimal balance of fitness and fatigue must be achieved to maximise (physical) performance. Balancing an athlete's training load with sufficient recovery and rest, aids in preventing excessive fatigue while also promoting adaptations and/or maintaining fitness.

This theory provides the foundation for load management decision making. Of course, we still have much to learn in this area. We need to better understand the causal pathways of injury and how the complex systems approach applies to sport.

In the applied setting however, we are responsible for applying current knowledge and technology to support decision making. So, what decisions are being made and what interventions are being employed?

Some fear a leap to pulling athletes out of training and games when it comes to load monitoring. How often does that actually happen? Is it reality or a widespread narrative that we are battling?

For me, pulling a player out of a game or even training on load monitoring grounds is the absolute last action taken.

It would only occur if supported by significant injury concerns and/or notably poor response data. Plus the decision would have to be made in a collaborative, interdisciplinary manner.

More often, we are working from a wider menu of interventions. See the diagram on the following page for a hierarchy of load management interventions.

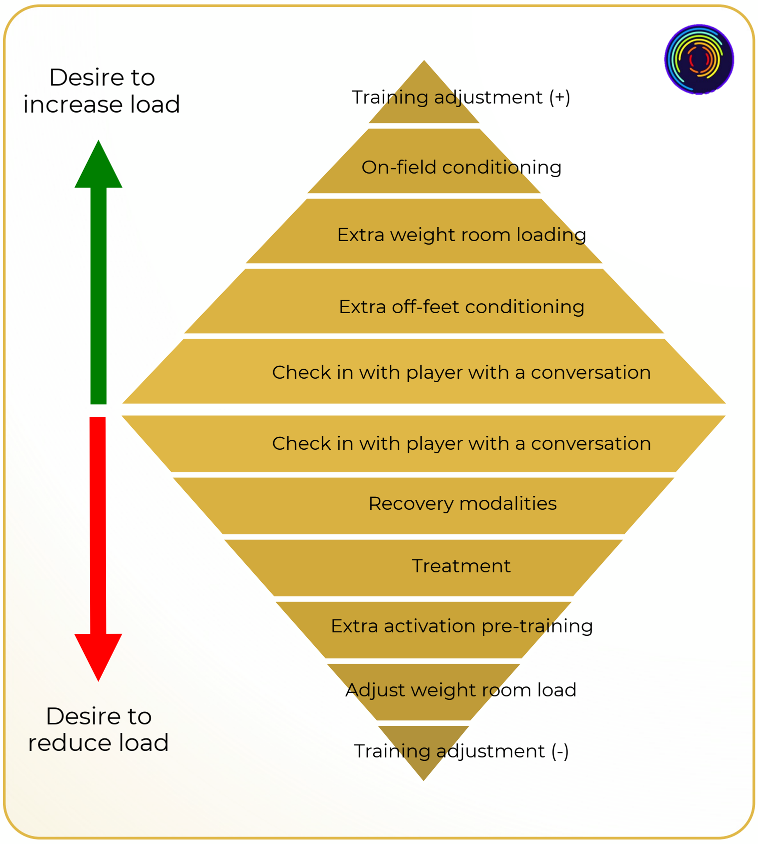
These interventions go in both directions; underloading and overloading. It is not just about reducing training load. There should be just as much debate about increasing workload. Some may shy away from this, as there seems to be a greater responsibility if something goes wrong. Yet, we know the benefits of building chronic workloads and physical capacities.

Man AND Machine: A Hybrid Approach

So, how might AI be integrated into this process?

As we discussed previously, the output design is crucial in guiding the user. AI applications outside of sports science may warrant a "go/no-go" decision. Yet, as the figure on the following page demonstrates, there are many potential layers of intervention in load management. Providers should design interpretable and actionable outputs, but this remains far from making the final decision in place of the human operator.

Application: *How do practitioners apply the outputs?*



In the book, The Signal and the Noise, Nate Silver talks extensively about a hybrid approach to decision making in sport. He advocates for scouts to use both intuition and algorithms.

“The fuel of any ranking system is information - and being able to look at both scouting and statistical information means that you have more fuel.”

A hybrid approach is also suited to training load management. While constantly evolving, sports science data inputs are not all encompassing. And although we’ve discussed the importance of layering contextual information into analytics, there is a multitude of day-to-day context that is only visible to the practitioner.

I suspect, regardless of AI, most teams and practitioners currently operate some form of hybrid approach, utilising a mixture of data and intuition to guide these decisions.

When it comes to the specifics of how to apply a hybrid approach, there is no one-size-fits-all approach.

Rich Buchanan
Zone7 Performance Director

How Zone7 Users Integrate a Hybrid Approach

Practitioners working in data-rich environments without an AI system are absolutely operating in a hybrid manner by conducting manual analysis of siloed datasets and cross referencing with expert human opinion. So, when introducing machine learning data analysis to this established hybrid approach, it is important for practitioners to fully understand what the added value really is.

A mistaken perspective I sometimes encounter in the industry, is one where there is a desire or expectation for AI to be definitive and flawless. To devolve responsibility to the machine, receive direction and instruction in pursuit of no false positives or false negative cases.

This, however, is unrealistic in a radically uncertain environment that is sports performance.

Application: *How do practitioners apply the outputs?*

Despite AI-derived insights being generated more efficiently, comprehensively and consistently compared to analogue methods, they still need to be cross-referenced with expert human knowledge. Once it is accepted that AI-derived insights are probabilistic and provide a positive contribution to a hybrid working model, then meaningful application and impact is far more likely.

Practitioners utilising Zone7 often find great comfort when the system reaffirms their own perspective, improving psychological certainty when determining the most appropriate course of action to take from the intervention hierarchy.

A greater challenge for practitioners is when the system presents information that conflicts with their professional perspective. However, it is in this situation that I personally feel the greatest value is to be gained from tools like Zone7.

A scenario where man and machine have conflicting perspectives in the workload management and injury risk forecasting domains should be viewed as a safety net. At the very least, it should facilitate further discussion and consideration.

Following further multidisciplinary discussion, it may change opinion and ultimately course of action. This is not a case of being data-driven by a machine, but simply man and machine operating as dependable teammates.

Dr Imtiaz Ahmad, Head of Medical at QPR FC, illustrated this for a recent masterclass:

"The way I see it is as part of our team. So, if I've got six or seven decision makers that work with me at QPR, I consider Zone7 as one of those as well. It is like having another member of staff – a member of staff that has got different opinions to me and comes to those [opinions] in different ways. Sometimes I like it, sometimes I don't."

Image: QPR's Dr. Imtiaz Ahmad consults with a member of his performance team.

Conclusion

Practically applying lessons from this report

Conclusion: *Practically applying lessons from this report*

Jo Clubb
Global Performance Insights

In this report, we have travelled through the entire data pipeline. From inputs to outputs, considering context and analysis, before discussing how this system can be implemented in the real world.

In the final chapter on implementation, we presented a hybrid approach that combines the strengths of both Man and Machine. This approach is echoed in Ed Smith's recent book, *'Making Decisions; putting the human back in the machine*'. The former England cricket selector surmised that the computer, algorithm, or AI, is simply an additional form of intelligence on which to call upon.

"Selection and decision-making are often framed in terms of 'art versus science', with the assumption that, in our digital age, 'science' is increasingly marginalising the human factor. But making decisions — and this applies in any area, not just sport — demands weighing and reconciling different kinds of information, and drawing on differing types of intelligence."

"No system, in other words, is so good that it can survive without good judgement. You can't box off a perfect process. Understanding the data can embolden better risk-taking, but it can't absolve decision-makers from responsibility."

No system, in other words, is so good that it can survive without good judgement. You can't box off a perfect process. Understanding the data can embolden better risk-taking, but it can't absolve decision-makers from responsibility.

Practical applications are of most interest to discerning practitioners. There are steps to take throughout the entire process to ensure an appropriate approach to load management with AI data. The checklist overleaf demonstrates the recommended Dos and Don'ts based on each step taken through the data pipeline in our recent series.

Conclusion: *Practically applying lessons from this report*

	DO	DON'T
Definitions	… use the AI Dictionary to expand your understanding of key terms in AI.	… worry about being unsure of terminology, AI is a rapidly evolving area and asking questions is expected.
Inputs	… conduct regular data diagnostics and maintain clean and consistent data practices with all inputs. … consider inputs beyond player tracking technology and challenge providers to incorporate such data. … review sports science data streams in relation to cadence, volume, and quality.	… forget the vital importance of clean data. Beware of Garbage In Garbage Out. Be a Data Steward. … change velocity thresholds or other settings without discussing modelling implications first. You should have a strong reason to change settings such as these, given the effect on historical datasets. … underestimate the importance of athlete buy-in to data quality (e.g., RPE/wellness questionnaire honesty)
Context	… incorporate your own knowledge, observations, and contextual information as important factors, both in analysis and application. … provide injury history, player profile (i.e. position, age, weight and height), and team periodisation information to be incorporated into the model … explore/ask providers how contextual factors influence the model (e.g., playing position, time in season etc.)	… dismiss what you know or see about a player or situation. Understanding of context is important. … forget that we humans are biased! Sometimes what we take to be "context" (e.g., "s/he is injury prone) might not be accurate.

Conclusion: *Practically applying lessons from this report*

	DO	DON'T
Analytics	... demand validation studies from prospective providers and request a custom validation study on your own data. ... vet the provider's protections against overfitting.	... settle for black boxes. Demand transparency and explainability.
Outputs	... discuss risk policies that are most suited to your team and have providers calibrate their models to reflect this. ... explore the underlying risk factors driving the outcomes. ...seek outputs that can expedite the training needs of athletes.	... accept every output as perfect. Question further, particularly those that surprise you. ... be afraid to provide feedback on data visualisation.
Application	... apply a hybrid approach that uses data with experience and intuition. ... consider the system as one more source of information/tool in the toolkit. ... consume information with a combination of curiosity and scepticism. ... keep a decision making diary so you can audit the decision making processes.	... surrender decision making entirely to the AI system. ... be led purely by an injury risk minimisation lens - sports performance requires some inherent level of risk.

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